



## Part I

### Lab Assignment-10 (Introduction to Object-Oriented Programming)

Department: CSE  
(22CPP209)

Subject: OOAD Lab

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**Objective:** Random number and Exception handling.

*Instruction:* Explore the use of function available with input stream class (like- fail(), ignore(), clear() etc.), read() and srand() functions. Students are advised to refer to cppreference page for detailed functionalities.

**Question 1.** Use a one-dimensional array to solve the following problem. Read in 20 numbers, each of which is between 10 and 100, inclusive. As each number is read, validate it and store it in the array only if it isn't a duplicate of a number already read. After reading all the values, display only the unique values that the user entered. Provide for the "worst case" in which all 20 numbers are different. Use the smallest possible array to solve this problem.

**Question 2:** Create a program that simulates drawing lottery numbers and checks them against a user's ticket, incorporating exception handling to manage input errors effectively. The program will prompt the user for the number of lottery numbers to draw (1-10) and their ticket numbers (up to 10). It will generate unique random lottery numbers from 1 to 50 and validate the user input, throwing exceptions for invalid entries, such as exceeding limits or non-integer values.

For example, entering 11 for the lottery numbers should produce an error stating the count must be between 1 and 10. Similarly, entering more than 10 ticket numbers should trigger an appropriate error message. Invalid input types, like letters instead of numbers, should prompt a request for a valid integer. The program should also handle cases where no ticket numbers are entered or when valid ticket numbers don't match any generated lottery numbers.